



Gradient or no gradient: Spatial hydrostatic pressure distributions in bilayer thin-film composite membranes

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ABSTRACT

Water recovery through pressure-driven bilayer thin-film composite (TFC) membranes remains an open problem, with critical uncertainty centered on its driving forces, namely the hydrostatic pressure *gradient*, yet systematic investigation remains limited. Here, we present a poromechanics framework that explicitly accounts for the distinct mechanical, structural, and transport properties, elucidating the interplay between flow-induced compaction and water transport in a TFC membrane governing the transport upper limit performance. Our approach naturally splits the strain energy into two regions: (i) a linear-response selective layer and (ii) a non-linear supporting one, mechanically capturing the strain-hardening behavior as the compacted support layer transitions into its bulk polymer state. While the hydrostatic pressure distribution in the selective layer merely scales upward with the increasing applied transmembrane pressure, the distribution in the support layer, however, contorts from a zero slope to a linear one and finally to a non-linear slope, demonstrating how the water transport changes to a support layer-controlled system. Overall, we show how a cancellation between permeance and the hydrostatic pressure difference across TFC membranes drives the transition between (i) support layer-controlled and (ii) selective layer-controlled regimes.

1. Introduction

Pressure-driven membrane technologies, particularly nanofiltration (NF) and reverse osmosis (RO), have become leading solutions for addressing water scarcity due to their comparatively efficient energy consumption in desalinating water resources [1–5]. At the core of these desalination processes are thin-film composite (TFC) membranes, which typically comprise two *primary* layers—(i) a selective layer, and (ii) a support layer—along with a *secondary* backing layer. The selective layer, with a thickness ranging from 15 to 300 nm [6,7], is composed of dense, crosslinked polyamide featuring sub-nanometer free volume elements that regulate water permeation and salt rejection [8,9], defining the permselectivity performance of these membranes. Adjacent, the support layer—an ultrafiltration membrane with a thickness of approximately 60 μm [10–12]—is typically made of polysulfone or polyethersulfone [13], providing a surface for interfacial polymerization [7] and structural integrity to the selective layer. Last, but not least important, the non-woven fabric-based backing layer provides further reinforcement, allowing these composite membranes to withstand high-pressure loading. While the secondary backing layer is often ignored, being assumed to have a negligible direct role in influencing the

selective layer's transport performance, the same cannot be said for the support layer, whose influence on water permeance through composite membranes under different mechanical cues, in our opinion, remains incompletely understood.

Despite extensive research on TFC membranes, the synergistic influence of the selective and support layers on the flow-induced compaction of the membranes remains unclear, particularly with regard to the spatial hydrostatic pressure distribution across these layers — which drives water transport — has largely been unexplored. As a result, optimizing performance-limiting factors, particularly the upper limit of water flow during operation, remains a persistent challenge. In operation, pressurizing the feed solutions of these membranes generates a transmembrane pressure (TMP), driving water flow through them and, concurrently, limiting water permeation as the membranes compact and densify, ultimately altering their morphology. Thus, this study is particularly important for high-pressure reverse osmosis (HPRO), where the operating pressure ($\gtrsim 100$ bar [1]) is substantially higher than conventional RO and NF systems. We, as well as others, identify three categories in the reporting of deformation in commercial TFC membranes, particularly in the selective and support layers: (1) larger strain in the selective

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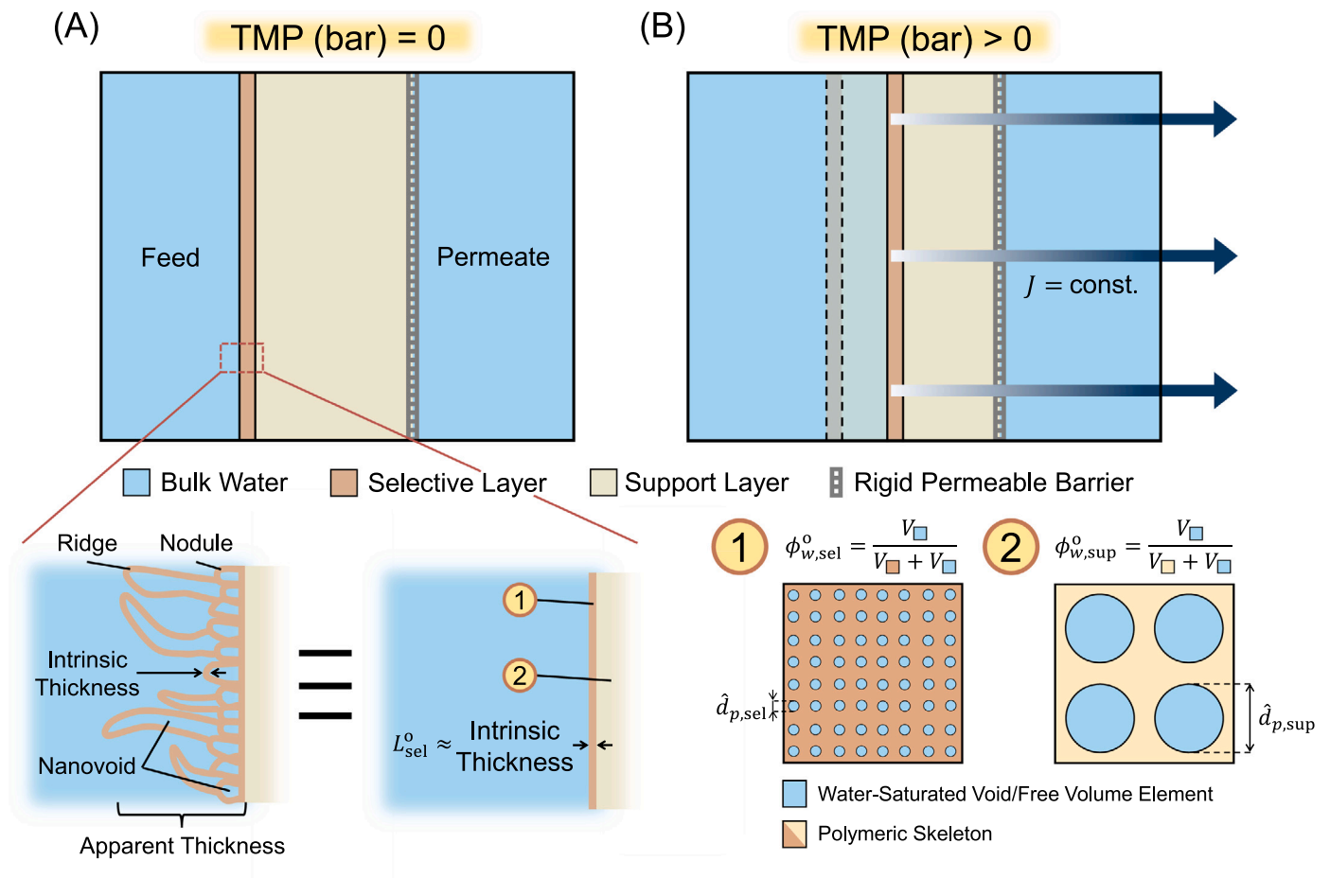


Fig. 1. Schematic representation of our model's thin-film composite (TFC) membrane (A) before and (B) after flow-driven compaction. Our TFC membrane comprises a bilayer structure (selective layer and support layer) sandwiched between feed and permeate water, and supported by a rigid permeable barrier at the membrane/permeate interface. Note that the selective and support layers are not drawn to scale due to the significant length difference (see SI: Table S1). The inset below (A) visualizes how we simplify the selective layer, characterized by a “ridge and valley” structure together with a denser nodular layer, into an equivalent selective layer with a thickness approximately equal to the intrinsic thickness (detailed explanation available in SI: Section S1). The inset below (B) provides an ideal illustration of the undeformed selective and support layers, both of which comprise water-saturated pores and a polymeric skeleton. The morphology of these layers is characterized by the average pore size, $\hat{d}_p$, and the initial porosity (or water volume fraction), ϕ_w^o . Mathematically, ϕ_w^o is defined as the total volume of pores divided by the total volume of the porous matrix (including both water and polymer).

layer [14], (2) greater strain in the support layer [13,15], and (3) significant strain in both layers [11,16]. These contrasting observations suggest that the deformation may be influenced by factors such as membrane material properties, operating conditions, and the specific interactions between the layers during pressure application, leading to variations in hydrostatic pressure gradients within them at different feed solution pressurization intensities.

To simulate water transport through TFC membranes, particle-based methods – especially molecular dynamics (MD) simulations at the atomistic level – are widely used to study the transport and compaction of the selective layer under varying crosslinking degrees, monomer compositions, and thicknesses [17–19]. Fully atomistic simulations, while detailed, require significant computational resources and cannot reach the time and length scales relevant to engineering applications. As a result, they are typically limited to analyzing the nanoscopic selective layer. This limitation prevents the observation of the plateau trend in the flux-pressure curve, which represents the upper limit of water flux, even when TMP is increased to 1500 bar [17–19], far exceeding current experimental TMP levels. On the other hand, theoretical modeling, such as viscoelastic or poromechanics theories, accommodates the engineering length and time scales required to study the deformation of the coupled selective and support layers, albeit sacrificing atomistic detail. To elaborate, existing viscoelastic models [10,20,21]

describe time-dependent compressive strain using spring–damper systems, where the components (springs and dampers) are arranged in a combination of series and parallel configurations. These models enable the prediction of real-time evolution of flux and *spatially-averaged* strain in pressure-driven membranes. Conversely, poromechanics models [22–28] excel in quantifying the spatial distributions of hydrostatic pressure and porosity within pressure-driven membranes. These models are fundamentally characterized by a set of continuum-based governing equations that couple fluid flow and compaction phenomena. Generally, existing poromechanics models exhibit variations in strain response and water transport mechanisms. Two notable recent examples include: (1) Fan et al. [28], who adopted a stress–strain response inspired by the Flory–Rehner (FR) theory, emphasizing thermodynamic water–polymer interactions, and (2) Hegde et al. [27], who employed a linear-elastic stress–strain response while integrating both pressure- and concentration-driven transport mechanisms into their model. However, we question the tractability and suitability of the published poromechanics methods: (i) Flory–Rehner’s theory for deriving the stress–strain response, where the mixing free energy generates osmotic pressure, resulting in non-realistically large compacting stress in drier membranes, as the water–polymer interactions dominate over the deformation mechanics of hydrophobic polymer chain [24,28–30] (see SI: Subsection S2.3); (ii) the omission of the strain-hardening/densification effect [27]; and (iii) the failure to consider the direct coupling between

the selective and support layers when determining water permeance, as recently pointed out [31].

Here we present a framework that combines the transport of water and the mechanics of SW30-XLE TFC membranes, resolving the spatial hydrostatic pressure distribution across their selective and support layers, i.e., the water transport driving forces, at different TMPs. We couple Darcy's law with our non-linear stress–stretch model, which captures the known yet often overlooked densification effects (due to the collapse of water-filled voids)—a material property change that membranes experience as they undergo severe (strain-hardening) deformation. While some models capture the exponential increase in compressive stress with compaction, their origin is rooted in thermodynamics rather than mechanics, leading to non-realistic flux behavior at high applied pressure (see SI: Subsection S2.4 for further discussion) [29]. Specifically, we aim to examine a specific scenario involving commercial TFC membranes, focusing on cases where the support layer is significantly more strained than the selective layer; otherwise, additional physics may need to be introduced. Although theoretical models/simulations have been developed for (point) zero-, one-, and even two-dimensional spaces, we adopt a 1D approach to conserve computational resources while mimicking experimental and simulation setups. This approach requires, at most, two categories of two property inputs for each membrane layer: (a) morphology—(i) porosities and (ii) pore sizes, and (b) stress–stretch—(iii) compressive moduli and (iv) strain-hardening factors. With these parameters, we can demonstrate how the hydrostatic pressure distribution changes in the membranes at various TMPs, transitioning from linear to non-linear behaviors.

2. Methodology

Our poromechanics framework elucidates the interplay between water transport and mechanical deformation within TFC membranes. We model the membranes as a bilayer system: (1) a permeation-controlling selective layer and (2) a porous support layer, in which each layer consists of two microphases: (a) polymeric skeleton and (b) water-saturated voids/free-volume elements. Applying TMP across the membranes causes water flow to compact them, reducing their thickness macroscopically and porosity microscopically, which together modulate their transport properties. To this end, our theory integrates non-linear frameworks for both (i) transport and (ii) mechanics to capture the interdependence between the membrane compaction and the decline in water permeability. Next, we explain the governing equations and then discuss the complementary boundary conditions.

2.1. Governing equations: Coupling between transport and mechanics

In this paper, we formulate our framework within the Lagrangian frame, a reference frame that is fixed in space. For simplicity, we coincide this reference frame with the pristine (undeformed) configuration of the TFC membrane. Consequently, the key space-dependent variables in our framework are nominal quantities, which are defined relative to the pristine (undeformed) configuration of the TFC membrane. To reduce our parameter space, we assume unidirectional (i) water flow and (ii) membrane deformation, with these and related quantities depending solely on the spatial coordinate in the thickness direction, denoted as, X .

We begin with water transport, modeled using the pore-flow approach, where a pressure gradient drives the water flow through the TFC membrane. Classically, the Hagen–Poiseuille equation, a representative model for pore-flow transport derived from the no-slip interaction between water and the pore wall, ideally describes the water flux through a straight pore [32]. In this work, we adopt Darcy's law, an expression analogous to the Hagen–Poiseuille equation that averages water flow over a volume containing numerous tortuous pores [32], to describe the water flux in our TFC membrane. Specifically, Darcy's law dictates that volumetric water flux (also known as the superficial

velocity) changes proportionally to the hydrostatic pressure gradient, written as

$$J = -\left(\frac{1}{\lambda_X}\right) \frac{k}{\eta} \frac{\partial P_h}{\partial X} \quad (1)$$

where k , η , λ_X , P_h and J are the Darcy's permeability, water viscosity, local stretch ratio (compressed infinitesimal length/initial infinitesimal length), hydrostatic pressure, and the superficial velocity (i.e., the steady-state volumetric water flux), respectively. Eq. (1) differs slightly from the standard form [33,34], with the correction term $1/\lambda_X$ accounting for the membrane's length change under compression. This correction, consistent with the one by Macminn et al. [26], effectively maps between the membrane's initial configuration to its true compressed one [35], reflecting that water permeation occurs within the compressed state.

To link the macroscopic flux, Eq. (1), with the microscopic properties, we allow the membrane's morphology to modulate the water permeation. Reasonably, a highly porous support layer characterized by bigger porosity and pore size should demonstrate superior water permeation compared to a denser selective layer. Hence, we express the Darcy's permeability as a function of porosity and pore size using a modified Carman–Kozeny expression (derivation in SI: Section S3), given as

$$k = \frac{\phi_w^3}{32} \hat{d}_p^2 \quad (2)$$

where ϕ_w , and $\hat{d}_p$ are void porosity (water volume fraction) and (effective) average pore size, respectively. We utilize Eq. (2) under the closed-packed spheres assumption [36], which inherently limits our ability to completely capture the morphological heterogeneity of the membrane. For example, the support layer comprises an irregular mixture of a (i) sponge-like dense region, alongside (ii) macrovoids (for instance, large finger-like pores) [11,13,21,37,38]. Significantly, finger-like pores in the support membrane would enhance membrane water permeation performance by reducing its local transport resistance compared to denser sponge-like regions [21]. In other words, under the same applied pressure and neglecting compaction effects, a support layer with a microstructure dominated by finger-like pores would demonstrate greater water flux than one primarily composed of sponge-like regions. Undoubtedly, for a morphologically heterogeneous TFC membrane, water flux at the molecular level varies sensitively with local changes in membrane microstructure. However, since this work highlights the upper limit of water permeation performance at the macroscopic level, it necessitates an analytically tractable approach that accounts for the overall morphological influence of the bilayer TFC membrane. To achieve this, Eq. (2) represents the morphology of the selective and support layers using just two locally averaged quantities: $\hat{d}_p$ and ϕ_w . For example, a support layer with abundant finger-like pores can be represented by a large $\hat{d}_p$ and ϕ_w , while a less porous support layer with a mostly sponge-like region can be assigned a smaller $\hat{d}_p$ and ϕ_w . This enables us to capture the complex heterogeneous morphology of selective and support layers by just using a single expression, seamlessly predicting the upper-limit permeation performance of the bilayer TFC membrane.

It is worth noting that, in addition to the morphology (e.g., pore size and porosity), other physicochemical factors, such as the composition and hydrophilicity, would indirectly influence the transport properties of the TFC membrane. For example, in the selective layer, adjusting the monomer concentrations for interfacial polymerization (IP) — typically *m*-phenylenediamine (MPD) and trimesoyl chloride (TMC) — would alter the polyamide crosslinking degree, ultimately affecting the water permeation performance [39,40]. Similarly, the support layer, when prepared under different conditions — such as varying polymer concentrations [38] and the inclusion of additives [41]

— can exhibit differences in hydrophilicity and surface pore size/porosity. These variations influence the formation of the polyamide layer during the IP process, ultimately affecting the overall performance of the TFC membrane. Nevertheless, since this work serves as a primer for investigating pressure distributions in pressure-driven desalination membranes, the consideration of these physicochemical properties is beyond the current work scope but could be explored in future research.

Next, we describe how our membranes respond under applied hydrostatic pressure, when water flows across them. Specifically, during steady-state operation, we assume that the TFC membrane equilibrates with the flow-induced compacting stress. Thus, based on the principle of free energy minimization [30,42,43], the compacting stress can be written as

$$\sigma_c = -\frac{\partial W}{\partial \lambda_X} \quad (3)$$

where σ_c and W are the compacting stress and the Helmholtz free energy density, respectively. Physically, the compacting stress results from the change of the hydrostatic pressure in the membranes, and a mechanical equilibrium expression governs their relationship, namely [26–28]

$$\frac{\partial P_h}{\partial X} + \frac{\partial \sigma_c}{\partial X} = 0. \quad (4)$$

The Helmholtz free energy, W , depends solely on a strain energy function, with flow-induced compaction decreasing the membrane thickness while increasing its strain energy. In other words, $\partial W / \partial \lambda_X$ characterizes the membrane's strain response toward the induced stress. To define the stress–stretch relationship (equation of state) for our bilayer membranes, we derive separate expressions for $\partial W / \partial \lambda_X$ for the selective and support layers, recognizing that their distinct morphologies and materials result in different deformation behaviors.

For the selective layer, MD studies indicate that the polyamide selective layer compresses linearly with TMP up to 1500 bar [17–19], justifying our use of a linear stress–stretch response to express $\partial W / \partial \lambda_X$, which can be written as

$$\frac{\partial W}{\partial \lambda_X} = K_{\text{sel}}(\lambda_X - 1), \quad \text{for } X_{\text{feed/sel}} < X < X_{\text{sel/sup}} \quad (5)$$

where K_{sel} denotes the selective layer's compressive modulus, while $X_{\text{feed/sel}}$ and $X_{\text{sel/sup}}$ are the nominal coordinates for the feed/selective layer and selective layer/support layer interfaces, respectively.

Conversely, experiments have shown that the support layer compresses less with increasing applied pressure [13,44,45], a behavior attributed to the densification of its porous matrix. Therefore, we phenomenologically propose an expression for its stretch response, written as

$$\frac{\partial W}{\partial \lambda_X} = \frac{K_{\text{sup}}^0}{\beta_1} \sinh[\beta_1(\lambda_X - 1)] + \frac{\phi_{w,\text{sup}}^0}{\beta_2} [K_{\text{sup}}^0 \cosh(\beta_1 \phi_{w,\text{sup}}^0) - K_{\text{sup}}^f] \left(\frac{1 - \lambda_X}{\phi_{w,\text{sup}}^0} \right)^{\beta_2}, \quad \text{for } X_{\text{sel/sup}} < X < X_{\text{sup/perm}} \quad (6)$$

where $X_{\text{sup/perm}}$ and $\phi_{w,\text{sup}}^0$ separately denote the support layer/permeate interface nominal coordinate and the support layer initial porosity, while K_{sup}^0 and K_{sup}^f are the support layer compressive moduli at pristine and fully-compacted states, respectively. In other words, K_{sup}^0 characterizes the stretch response of a water-saturated membrane with uncompacted polymer network, while K_{sup}^f describes the one for solid bulk polymer material. Notably, β_1 and β_2 numerically control the degree of strain-hardening caused by the densification of the support layer in the moderate and big compressive strain regimes, respectively. We direct readers to SI: Section S4 for a detailed derivation of Eq. (6) alongside the influence of each control parameter.

Having established the governing equations for the deformation mechanics and the water transport, we now couple these phenomena within our poromechanics model by utilizing an incompressibility constraint. By assuming a water-saturated pore space and neglecting the moisture effect, we can effectively enforce the incompressibility constraint by considering the finite volume occupied by water molecules and the polymeric skeleton in each layer (derivation presented in SI: Section S5), namely

$$\lambda_X = 1 - \phi_w^0 + \rho_w v_w \quad (7)$$

where ρ_w is the nominal water density (number per undeformed membrane volume) and v_w is the water molecular volume. Altogether, $1 - \phi_w^0$ and $\rho_w v_w$ signify the nominal volume fraction (current component volume divided by undeformed membrane volume) of the polymeric skeleton and water, respectively. As a limiting case, Eq. (7) reduces to $\rho_w v_w = \phi_w^0$ for the porous matrix in the undeformed configuration ($\lambda_X = 1$). This recovery of the water volume fraction to the initial porosity reflects our water-saturated pore assumption.

To reduce the degree of freedom in our framework, the nominal water density, ρ_w , is related to the true porosity, ϕ_w , through the following expression

$$\phi_w = \frac{\rho_w v_w}{\lambda_X}. \quad (8)$$

Accordingly, Eqs. (7) and (8) serve as a mathematical linkage between the water permeation (dictated by the water volume fraction) and the porous matrix deformation (characterized by the stretch ratio), thereby reducing the parameter space of our poromechanics model.

2.2. Boundary conditions

We impose three boundary conditions at the interfaces to acquire physically consistent numerical solutions within our computational domain (the selective and support layers). Firstly, we set the hydrostatic pressure at the support layer/permeate interface to the atmospheric level, namely

$$P_h(X_{\text{sup/perm}}) = 0. \quad (9)$$

Secondly, we assume continuity of hydrostatic pressure at the selective/support interface, given as

$$P_h(X_{\text{sel/sup}}^-) = P_h(X_{\text{sel/sup}}^+) \quad (10)$$

where $X_{\text{sel/sup}}^-$ and $X_{\text{sel/sup}}^+$ represent the nominal locations marking the end of the selective layer and the beginning of the support layer, respectively.

Thirdly and lastly, we assume zero compaction stress at the feed/selective layer interface, expressed as

$$\sigma_c(X_{\text{feed/sel}}) = 0. \quad (11)$$

In summary, Eqs. (1), (3), and (4), along with the constitutive Eqs. (2), (5), and (6), form the core of our poromechanics framework, collectively enabling us to describe the coupled deformation behavior and water transport characteristics of TFC membranes. We connect the mechanical and transport parameters via the relationship between water nominal density, porosity, and stretch ratio, defined by Eqs. (7) and (8), ensuring proper constraint of all degrees of freedom. By incorporating the boundary and interface conditions (Eqs. (9), (10), and (11)), we obtain particular solutions that comprehensively describe the behavior of these membranes.

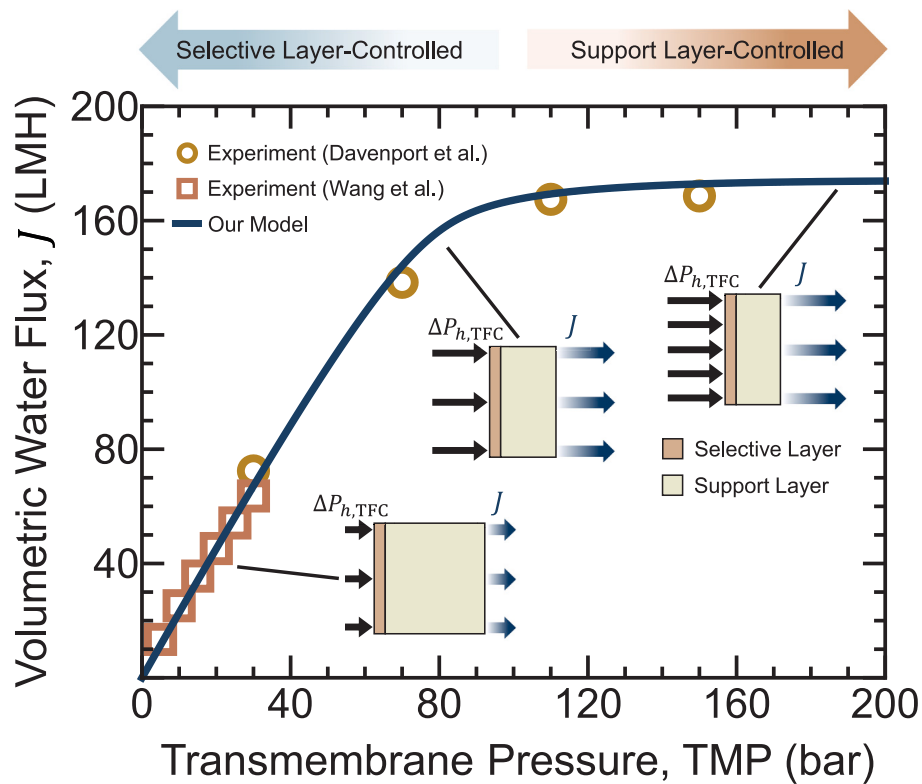


Fig. 2. Our predicted volumetric water flux across the TFC membrane, together with published experimental results [13,46], as a function of transmembrane pressure (TMP). Herein, TMP is also denoted as $\Delta P_{h,TFC}$. The blue 'Selective Layer-Controlled' arrow (pointing left) indicates the linear flux–pressure relationship in the low TMP regime, while the brown 'Support Layer-Controlled' arrow (pointing right) represents the plateauing flux trend in the high TMP regime. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Results and discussion

3.1. Validating the upper limit of water flux

To understand the transport performance limits of pressure-operated TFC membranes, we quantify their permeation performance by predicting water flux under increasing applied pressures, demonstrating the transition from a (i) selective-layer-controlled system to one governed by the (ii) support layer. Experimentally, the relationship between water flux, J , and TMP, $\Delta P_{h,TFC}$, is commonly expressed as $J = A_{TFC} \Delta P_{h,TFC}$ where A_{TFC} represents water permeance of the TFC membrane, typically measured in $L \cdot m^{-2} \cdot h^{-1} \cdot bar^{-1}$ or LMH/bar. In other words, the steady-state water flux intensity across the membranes is governed by the product of (i) the TMP and (ii) the experimental water permeance. However, the subtle interplay between them complicates the quantification of water flux due to the presence of a supporting layer (explained further in the next subsection). Here, we first validate the predictive capacity of our poromechanics framework by illustrating the correlation between TMP and volumetric flux.

Fig. 2 shows our predicted relationship between the steady-state volumetric flux, J , and the TMP, $\Delta P_{h,TFC}$, through solving Eqs. (1), (3), and (4). Notably, we specify our framework to the characteristics of the SW30-XLE TFC membranes, a seawater RO membrane produced by Dupont Water Solutions. This enables us to validate our results against the experimental data reported by Davenport et al. [13] and Wang et al. [46]. Although our results are not validated for other classes of pressure-driven membranes, specifically NF membranes, we anticipate that RO and NF membranes share key similarities in their operating mechanisms. In practice, RO membranes exhibit an order of magnitude lower permeance than NF membranes [40], which can be attributed to their significantly smaller pore size [47,48]. However, water flow in both RO and NF membranes is commonly driven

by hydrostatic pressure gradient, where changes in water pressure lead to flow-induced compaction. Therefore, in our opinion, we think that examining RO membranes alone is adequate to representatively highlight the upper-limit permeation performance of TFC membranes. Our computed results agree well with the experimental data because we capture the membrane's physical properties, such as dimensions, morphology, and structural characteristics, within our framework. We describe these properties mathematically using reasonable numerical inputs, including layer length, pore size, porosity, and stress–stretch response. For a detailed overview of the control variables and their corresponding values used in this study, refer to SI: Table S1.

Let us discuss how volumetric water flux changes with increasing TMP, as this is a key issue in evaluating the validity of the solution-diffusion (concentration gradient-driven) and pore-flow (pressure gradient-driven) models for understanding water transport through pressure-driven membranes. We address this by coupling pore-flow transport with non-linear mechanics, which allows us to capture the upper limit of water flux—a feature commonly attributed to the solution-diffusion model (discussed further in Section 3.3). This coupling approach also explicitly accounts for the supporting layer in addition to the selective layer, unlike (i) existing poromechanics frameworks [22–28], where only a generic monolayer membrane is considered, and (ii) published MD simulation studies [2,17–19,46,49], where the selective layer is typically supported implicitly (as recently highlighted by Freger and Ramon [31]). The flux–pressure relationship shown in Fig. 2 follows a sigmoidal/bilinear trend characterized by two distinct regimes: linear and plateau.

Now, we describe the two limiting cases of water transport in TFC membranes in greater detail, namely the selective layer-controlled and support layer-controlled regimes, as illustrated in Fig. 2. In the low-pressure regime, the flux increases linearly with TMP, where water

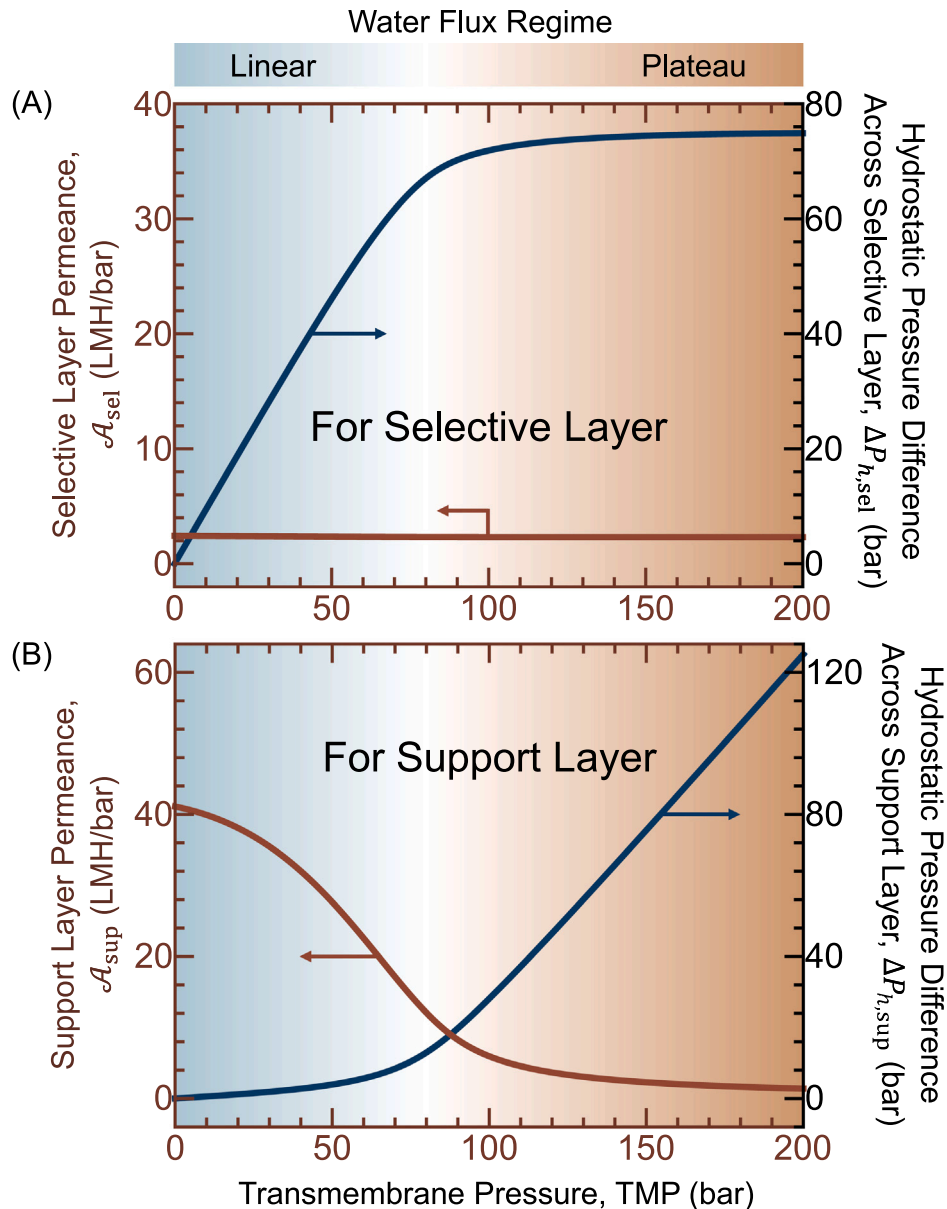


Fig. 3. (A) Selective layer permeance (dark blue line) and hydrostatic pressure difference across the selective layer (brown line), as a function of transmembrane pressure (TMP). (B) Support layer permeance (dark blue line), and hydrostatic pressure difference across the support layer (brown line), as a function of TMP. To align with Fig. 2, namely the flux-TMP curve, the low TMP regime (associated with linear flux-pressure relationship) is colored in blue, while the high TMP regime (corresponds to plateauing flux-pressure relationship) is colored in brown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

permeation is controlled by the selective layer, which has a substantially lower permeance compared to the support layer (see Fig. 3). This linear flux-pressure relationship in our work indicates an ideal operating regime where pressure-driven water transport through the TFC membrane is largely unhindered by the membrane compaction. However, as TMP continues to rise, the flux increment becomes noticeably lesser, due to the significantly elevated transport resistance associated with support layer compaction. Within our pore-flow-based poromechanics framework, this decreasing flux increment implies that a larger pressure gradient, and thus a higher TMP, is required to drive the water transport through the TFC membrane. Eventually, in the high-pressure regime ($\gtrsim 150$ bar), the flux increment becomes very miniscule compared to that in the selective layer-controlled regime, rendering the flux practically independent of TMP. At this point, the system transitions to being controlled by the support layer, which has a lower permeance than the selective layer (see Fig. 3). This flux leveling

behavior contrasts with the classical pore-flow model, which neglects compaction. However, modified variants, such as [22,24,26,27] and our present work, have addressed this limitation.

As a highlight, our model further predicts a material-dependent maximum operating TMP of ~ 310 bar for the SW30-XLE membrane (see SI: Section S6), corresponding to the compacting stress at which the support layer becomes fully-compacted. At this point, the support layer material, particularly at the support layer/permeate interface, transitions from a mixture of polymer and water to its bulk polymer state (i.e., $\phi_w = 0$), effectively blocking water passage through the TFC membrane.

Through Fig. 3, we investigate the individual permeances, $\mathcal{A}_i$, of the support and selective layers, which are generally expressed as

$$\mathcal{A}_i = \frac{J}{\Delta P_{h,i}} \quad (12)$$

where $\Delta P_{h,i}$ is the pressure difference across the (selective or support) layer, where $i \in \text{sup}$ (support) and sel (selective). In our framework, J and $\Delta P_{h,i}$ vary with TMP, as indicated by the non-idealized Darcy's law (see Eq. (1)). Specifically, by solving Eq. (1), alongside Eqs. (3) and (4), we could obtain the hydrostatic pressure distribution across the selective and support layers for an arbitrary water flux (J) value. This allows us to determine the pressure difference across each layer, namely $\Delta P_{h,\text{sel}} = P_h(X_{\text{feed}/\text{sel}}) - P_h(X_{\text{sel}/\text{sup}})$ for the selective layer, and $\Delta P_{h,\text{sup}} = P_h(X_{\text{sel}/\text{sup}}) - P_h(X_{\text{sup}/\text{perm}})$ for the support layer, where $X_{\text{feed}/\text{sel}}$, $X_{\text{sel}/\text{sup}}$, and $X_{\text{sup}/\text{perm}}$ denote the spatial positions at the feed/selective, selective/support, and support/permeate interfaces, respectively. Furthermore, the TMP can be written as $\Delta P_{h,\text{TFC}} = P_h(X_{\text{feed}/\text{sel}}) - P_h(X_{\text{sup}/\text{perm}})$. Altogether, we could extract a consistent set of J , $\Delta P_{h,i}$, and $\Delta P_{h,\text{TFC}}$ values from the mathematical solution of Eq. (1), enabling us to plot the permeance-TMP curve in Fig. 3.

Now, let us guide the reader in interpreting Fig. 3 by elaborating on two instances with different TMP values.

- When TMP = 50 bar (linear flux regime), the selective and support layers exhibit permeances of 2.36 LMH/bar and 27.63 LMH/bar, respectively, with corresponding pressure differences of 46.13 bar and 3.94 bar. Collectively, the product of permeance and pressure difference for each layer leads to a common steady-state flux of 109 LMH.
- When TMP = 200 bar (plateauing flux regime), the selective and support layers exhibit permeances of 2.32 LMH/bar and 1.39 LMH/bar, respectively, with corresponding pressure differences of 74.89 bar and 125.11 bar. Collectively, the product of permeance and pressure difference for each layer leads to a common steady-state flux of 174 LMH.

In Fig. 3(A), the selective layer's permeance, $\mathcal{A}_{\text{sel}}$, remains nearly invariant (2.32 ~ 2.43 LMH/bar) across the TMP range (0–200 bar). This implies that, intrinsically, the selective layer experiences minuscule compaction-induced permeation decline, as it possesses a GPa-order stiffness (see Table S1). Moreover, the dense selective layer possesses significantly lower permeance values than its support counterpart, especially in the low TMP regime. This demonstrates the permeation-controlling role of the selective layer in the low TMP regime, as depicted by the linear flux-pressure regime in Fig. 2.

In contrast, Fig. 3(B) shows that the support layer's water permeation decreases with TMP, where $\mathcal{A}_{\text{sup}}$ declines sigmoidally from 41 LMH/bar at pristine state, to about 1.4 LMH/bar at TMP = 200 bar, thus driving the flux plateau trend (support layer-controlled regime) shown in Fig. 2. As a result, to drive water transport through the support layer at high TMP, a higher hydrostatic pressure difference across the support layer, $\Delta P_{h,\text{sup}}$, is needed to balance the diminished water permeance, as dictated by Eq. (12). This demand causes the pressure distribution to follow a steep declining trend (which will be explained following subsection), satisfying the higher pressure difference requirement. As shown in Fig. 3(B), $\Delta P_{h,\text{sup}}$ stays mostly constant for TMP below 80 bar. However, for TMP above 80 bar, $\Delta P_{h,\text{sup}}$ increases linearly with TMP. In other words, the support layer consumes a substantial portion of the applied pressure within the high TMP regime.

Furthermore, the plummet of support layer permeance also creates a bottleneck for the overall water permeation performance of the TFC membrane system. To demonstrate, $\Delta P_{h,\text{sel}}$ in Fig. 3(A) increases linearly with TMP for values below 80 bar but plateaus thereafter, which indicates that the water-driving force in the selective layer is affected by the compaction-induced flux deterioration originating from the support layer. Collectively, the transitions of $\Delta P_{h,i}$ and $\mathcal{A}_i$ in Figs. 3(A) and (B) illustrate how differently these two layers navigate the differing TMP values, providing insights into the interdependence in permeation performance between the selective and support layers.

Notably, our results in Figs. 2 and 3 are consistent with both theoretical [24,26,27,50–52] and experimental [13,37,52,53] studies

on pressure-driven membranes, which emphasize that flow-driven compaction causes the flux plateau phenomenon. Under high TMP, severe compaction leads to significant microstructural and macroscopic material deformation in the membranes. As a result, permeability declines, causing only a negligible increase in flux despite the rise in TMP. This offsetting effect, where the decrease in permeability counteracts the increase in TMP, will be further elucidated in the following subsection, where we explore how increased TMP enhances the support membrane water transport resistance.

3.2. Support layer: How the linearity of hydrostatic pressure gradient changes with increasing (inverse of permeance) transport resistance

Here, we first quantify the overall TFC membrane permeance, $\mathcal{A}_{\text{TFC}}$, a prefactor that phenomenologically relates water flux intensity to TMP. We note that the inverse of the permeance represents its transport resistance [15], namely, $\mathcal{A}_{\text{TFC}} = 1/R_{\text{TFC}}$, where we demonstrate how the permeance decreases non-linearly with increasing TMP. Fig. 4, derived from the flux and applied pressure in Fig. 2, shows a minimal decline in $\mathcal{A}_{\text{TFC}}$ with TMP increases up to ~ 80 bar, but $\mathcal{A}_{\text{TFC}}$ significantly decreases as TMP increases further. Notably, the two distinct trends in the $\mathcal{A}_{\text{TFC}}$ behavior – the minimal and substantial declines – correspond to the linear and plateauing water flux regimes, respectively, as seen in the flux-pressure relationship in Fig. 2. Altogether, we can see that the flow is phenomenologically constrained by the permeance, $\mathcal{A}_{\text{TFC}}$. As a reminder to the reader, the water flux (J) is governed by the product of the permeance ($\mathcal{A}_{\text{TFC}}$) and the pressure difference across the TFC membrane ($\Delta P_{h,\text{TFC}}$), as expressed by $J = \mathcal{A}_{\text{TFC}} \Delta P_{h,\text{TFC}}$. While water flux is generally expected to increase with TMP, this expectation does not always hold, particularly at higher TMP values. As operating conditions transition from low to high TMP, the hydrostatic pressure difference across the TFC membrane does indeed increase; however, the significant decline in permeance accompanying this pressure increase, especially the support layer, restricts the rise in flux, ultimately resulting in a ceiling effect on water flux behavior.

Next, we unveil the spatial distribution of hydrostatic pressure across the TFC membranes at increasing TMP to further investigate the decline in the permeance. Figs. 5(A)–(D) illustrate a monotonic decrease in hydrostatic pressure toward the permeate side, with water molecules moving down these slopes in accordance with Darcy's law (Eq. (1)). Although the pressure drop within the selective layer is linear, the pressure drop in the support layer can exhibit more complex, non-linear behaviors, which ultimately govern the *global* hydrostatic pressure distribution, considering the spatial distance across the selective and support layers. At high TMP, these non-linear pressure trends arise to overcome the significant support layer transport resistance R_{sup} due to the tight coupling between water transport and compaction mechanics at the local support layer level. As such, the support layer with elevated transport resistance demands a greater driving force for an increased water transport, which is realized by the steeper support layer hydrostatic pressure distribution, as visualized in Figs. 5(C) and (D).

To further emphasize the importance of the support layer, we quantify its contribution to the membranes' transport resistance by adopting a commonly employed resistance-in-series assumption [15,54–56]. We calculate the resistances of the selective and support layers, R_{sel} and R_{sup} , respectively, defined as the pressure drop across each layer per unit flux, following Aghajani et al. [15]. Based on this assumption, the total transport resistance, R_{TFC} , is the sum of the resistances in the selective and support layers: $R_{\text{TFC}} = R_{\text{sel}} + R_{\text{sup}}$. Note that, for simplicity, we neglect water entrance resistance at the selective/support interface. This simplification is justified by our finding that, under high TMP conditions, the compacting stress at the support/permeate interface is significantly higher than at the selective/support interface (see SI: Figs. S13 and S14). Thus, the critical point, i.e., the point with highest local transport resistance, lies at the support/permeate interface, rather

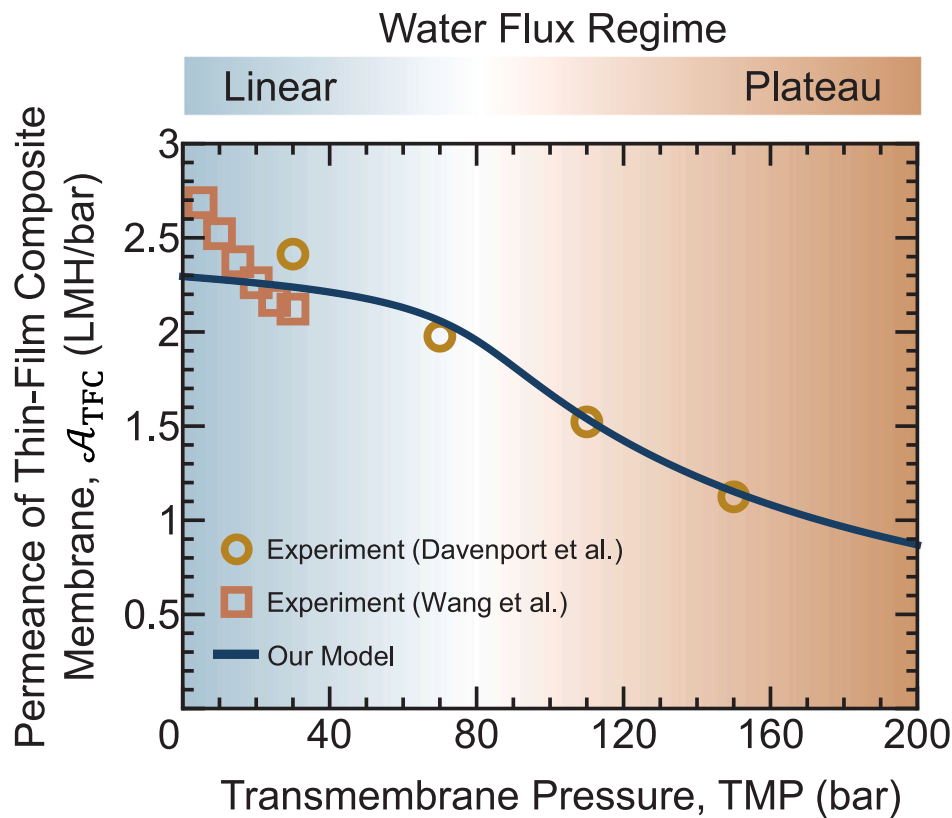


Fig. 4. The calculated thin-film composite (TFC) membrane permeance versus experimental results [13,46], as a function of TMP. To connect with Fig. 2, i.e., the flux-TMP curve, the low TMP regime (associated with linear flux-pressure relationship) is colored in blue, while the high TMP regime (corresponds to plateauing flux-pressure relationship) is colored in brown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

than at the selective/support interface. Mathematically, $R_{TFC} = 1/A_{TFC}$, which characterizes the hydrostatic pressure drop per unit flux in the *entire* TFC membrane. As a ratio, we visualize the normalized support layer resistance trend, namely R_{sup}/R_{TFC} , at different TMPs, as illustrated in Fig. 6, highlighting the support layer's contribution to the total transport resistance.

Consistent with the A_{TFC} trend in Fig. 4 and the A_{sup} one in Fig. 3, R_{sup} remains relatively low in the linear flux regime ($TMP \lesssim 60$ bar). As TMP further increases, R_{sup} rises substantially, indicating a significant impact on water transport, meaningfully modulating the flux as it plateaus at higher TMP values. Our results in Fig. 6 qualitatively align with those reported by Aghajani et al. [15]. Here, we observe that even with two distinct membranes, both indicate a monotonic increase in their support layer resistance with larger TMP values. Specifically, the maximum values of R_{sup}/R_{TFC} are comparable: Aghajani et al. [15] reported a maximum of 31% (calculated from their result, $R_{sup}/R_{sel} = 45\%$, based on the resistance-in-series equation, $R_{TFC} = R_{sel} + R_{sup}$) at 83 bar, while our framework predicts 63% at the highest investigated TMP of 200 bar, with our 31% point predicted at 105 bar. Under operating conditions typical of high-pressure reverse osmosis (HPRO) systems, which usually operate around this pressure level, the transport resistance of the support layer becomes significant and cannot be ignored.

When $R_{sup}/R_{TFC} \lesssim 0.1$ (the linear flux regime), the selective layer contributes the majority of the transport resistance, requiring a higher hydrostatic pressure gradient – i.e., driving forces – to transport water molecules through it compared to the support layer. When $R_{sup}/R_{TFC} \gtrsim 0.4$ (the plateauing flux regime), the support layer, on the other hand, contributes substantially to the total transport resistance, and its hydrostatic pressure gradient becomes more non-linear to facilitate water flow. The selective layer's hydrostatic pressure distribution remains relatively *linear* as TMP changes, shifting only up or down.

To this end, maintaining a constant flux through the TFC membrane requires each layer to adjust its permeance in response to its hydrostatic pressure difference, ensuring that the product (water flux) remains consistent across the layers. When the applied pressure increases in the high TMP zone, the support layer imposes the dominant resistance to the overall water transport, with little to no improvement in its flux intensity. Consequently, this resistance affects the driving force within the selective layer, resulting in minimal adjustments to its hydrostatic pressure gradient to maintain a constant flux through the membrane system, thereby compensating for the slight flux changes in the support layer. Collectively, our poromechanics framework demonstrates the interplay between the transport resistance and driving forces of the support and selective layers, highlighting how changes in the support layer extrinsically influence the transport performance of the selective layer. To further understand this finding, the following two subsections will investigate how pressure-driven flow across the TFC membranes alters their thickness as well as morphology, concomitantly impacting their local permeability.

3.3. How transmembrane pressure impacts TFC membrane porosity and local transport properties

This subsection examines how flow-induced compaction alters the membrane's void porosity, Φ , from the selective layer to the support layer and how these local structural changes affect water transport properties. For simplicity, we assume that Φ also represents the water volume fraction, ϕ_w , considering all voids or pores to be water-saturated (i.e., $\Phi = \phi_w$). Here, we characterize local water transport using Darcy's permeability, k , as defined in Eq. (2), which depends on the porosity ϕ_w .

Pressure-driven flow under TMP compacts the TFC membranes which leads to flux deterioration, and our framework quantifies this

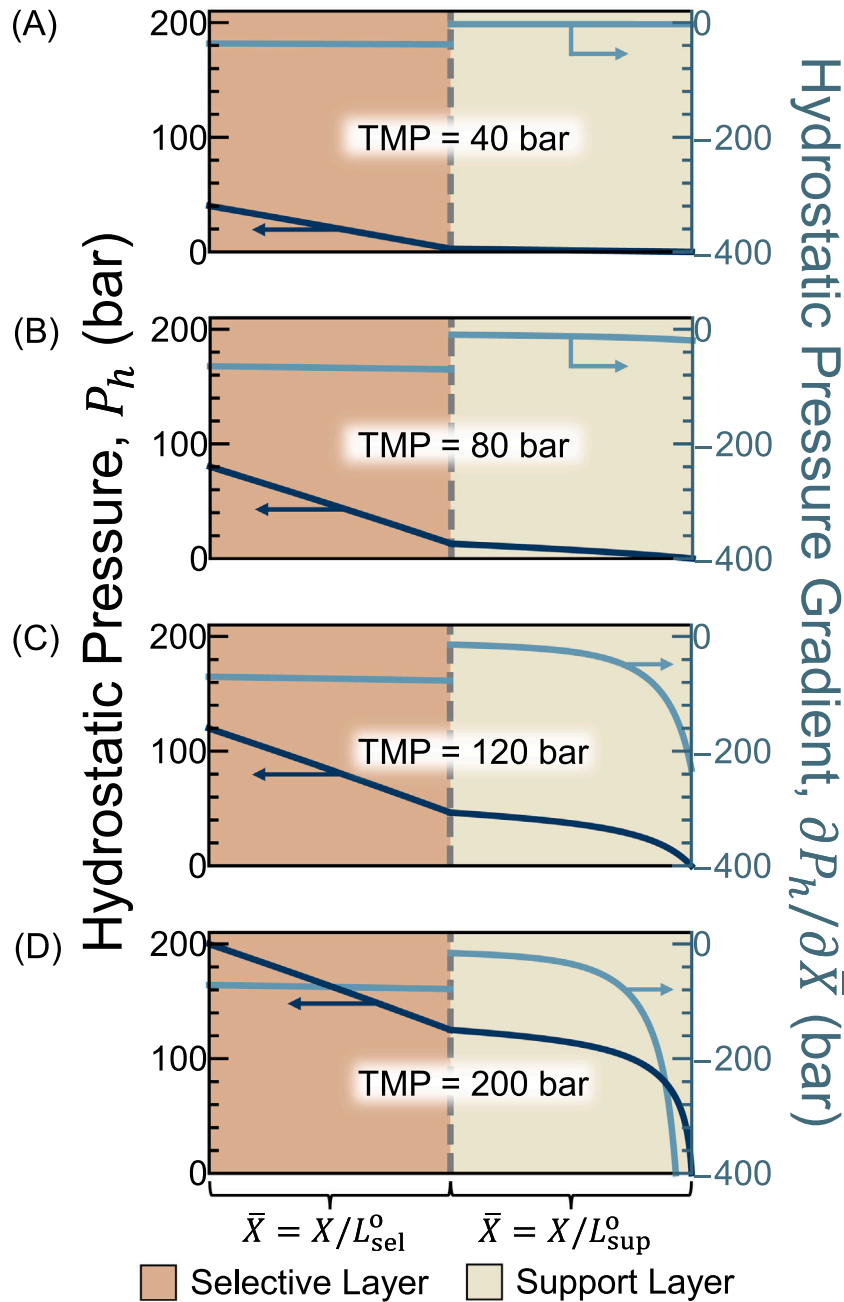


Fig. 5. The hydrostatic pressure, and its gradient, as a function of thickness-normalized nominal distance (relative to their initial lengths), at TMP of (A) 40 bar, (B) 80 bar, (C) 120 bar, and (D) 200 bar. Note that we adopt nominal coordinates, which is characterized by the membrane's undeformed configuration, to present the spatially dependent quantities in (A)–(D). Unlike true coordinates, which incorporate local thickness variations due to membrane compaction, nominal coordinates allow us to systematically compare pressure distributions at different TMPs by eliminating the influence of thickness changes, as demonstrated in SI: Section S7. This facilitates a straightforward understanding of pressure drop development within the TFC membrane at various TMPs.

compaction through changes in their length and porosity, demonstrating the tight coupling between water transport and membrane mechanics. To highlight the macroscopic effect of membrane compaction, we re-cast all space-dependent quantities in true spatial coordinates in Fig. 7, contrasting them with the nominal spatial coordinates used in Fig. 5. Specifically, we plot the true spatial distribution of porosity, $\phi_w(x)$, within the compacted membranes alongside the corresponding hydrostatic pressure distribution, $P_h(x)$. In solid mechanics, true coordinates (x), also known as the deformed configuration, capture the local changes caused by membrane compaction and accurately represent the membrane in its compacted state. In contrast, nominal coordinates (X) represent the spatial positions in the undeformed membrane. The

transformation from nominal to true coordinates can be obtained by solving the following expression, namely

$$\frac{\partial x}{\partial X} = \lambda_X. \quad (13)$$

The intrinsic/local stretch, λ_X , represents the ratio of the membrane's infinitesimal length after compaction to its original one, which is related to the water volume fraction, ϕ_w , through Eqs. (7) and (8). Mathematically, Eq. (13) is constrained by a boundary condition given by

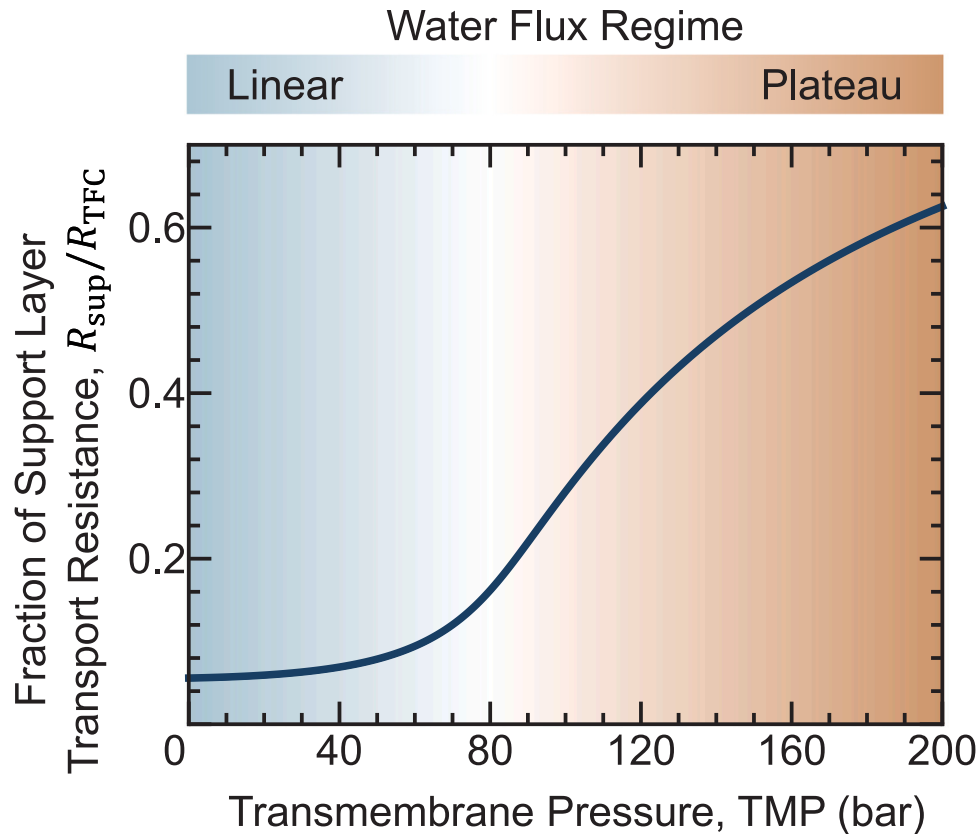


Fig. 6. The fraction of support layer resistance in the TFC membrane as a function of TMP. To connect with Fig. 2, specifically the flux-TMP curve, the low TMP regime (associated with linear flux-pressure relationship) is colored in blue, while the high TMP regime (corresponds to plateauing flux-pressure relationship) is colored in brown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$x(X_{\text{sup/perm}}) = X_{\text{sup/perm}} \quad (14)$$

where $X_{\text{sup/perm}}$ denotes the nominal coordinate at the support/permeate interface. Since the support layer is fixed to a fictitious rigid permeable barrier at this interface (see Fig. 1), Eq. (14) ensures that no displacement occurs at $X_{\text{sup/perm}}$.

When TMP = 0 bar, the membrane is in its pristine, undeformed state, with uniform but distinct initial porosities in both layers, as shown in Fig. 7(A). The initial porosities of the selective and support layers are defined as $\phi_{w,\text{sel}}^0 = 0.30$ and $\phi_{w,\text{sup}}^0 = 0.70$, respectively (see SI: Table S1), based on published experimental measurements and simulation data. These values are justified as follows:

- (i) For fully-aromatic selective layers, experimental water volume fraction results based on water uptake measurements range from 0.15 to 0.32 [57,58], suggesting that our SW30-XLE membrane, which also features a fully-aromatic polyamide layer [59], likely shares a similar chemical composition, physical morphology, and thus porosity. Additionally, a molecular dynamics simulation for the selective polyamide layer also reports porosities ranging from 0.24 to 0.31 in the pristine state [17].
- (ii) For the support layer, our initial porosity of 0.70 aligns with the total porosity ($\phi_t = 0.64$) reported in the Monte Carlo simulation by Xu et al. [21] for the SW30-XLE membrane. This value also falls within the typical range of 0.60–0.90 for support layer porosity, as reported in experimental studies [11,13,38,60,61]. These experimental porosity values are typically estimated using methods such as liquid-liquid porosimetry [13], water uptake

measurements [11], or software analysis of cross-sectional SEM images [61].

The pore-flow framework requires a driving force in the form of a hydrostatic pressure gradient, $\partial P_h / \partial x$, to allow water molecules to permeate the membranes. As a result of mechanics, a TMP gradually converts into internal compacting stress along the water flow direction (see Eq. (4) and SI: Section S7), increasing the membrane's strain energy, which alters the stretch ratio, reduces local porosity, and ultimately decreases the water permeability. The realization of this pressure transfer mechanism necessitates the implementation of zero-value boundary conditions for the hydrostatic pressure and compacting stress at the support layer/permeate and feed/selective layer interfaces, respectively (see Eqs. (9) and (11)). Consequently, the TMP *transmission* initiates at the feed/selective layer interface and culminates at the support layer/permeate interface, placing the membranes within the confines of the feed and permeate solutions.

When TMP = 50 bar, the support layer begins to compact noticeably, as evident from the comparison between Figs. 7(A) and 7(B). However, the porosity of the selective layer remains relatively unchanged, while that of the support layer decreases significantly (from 0.7 to 0.5), primarily because the selective layer is much stiffer than the support layer. The selective layer, with an elastic modulus in the GPa range [62,63], is 1–2 orders of magnitude stiffer than the support layer [11,38,64–66] (see SI: Section S8 and Table S1). Here, the denser selective layer, characterized by comparatively high stiffness and transport resistance, experiences a remarkable hydrostatic pressure drop. Through our mechanical equilibrium condition (Eq. (4)) and pressure continuity assumption (Eq. (10)), this pressure drop is then converted into a compacting stress that is exerted onto the support

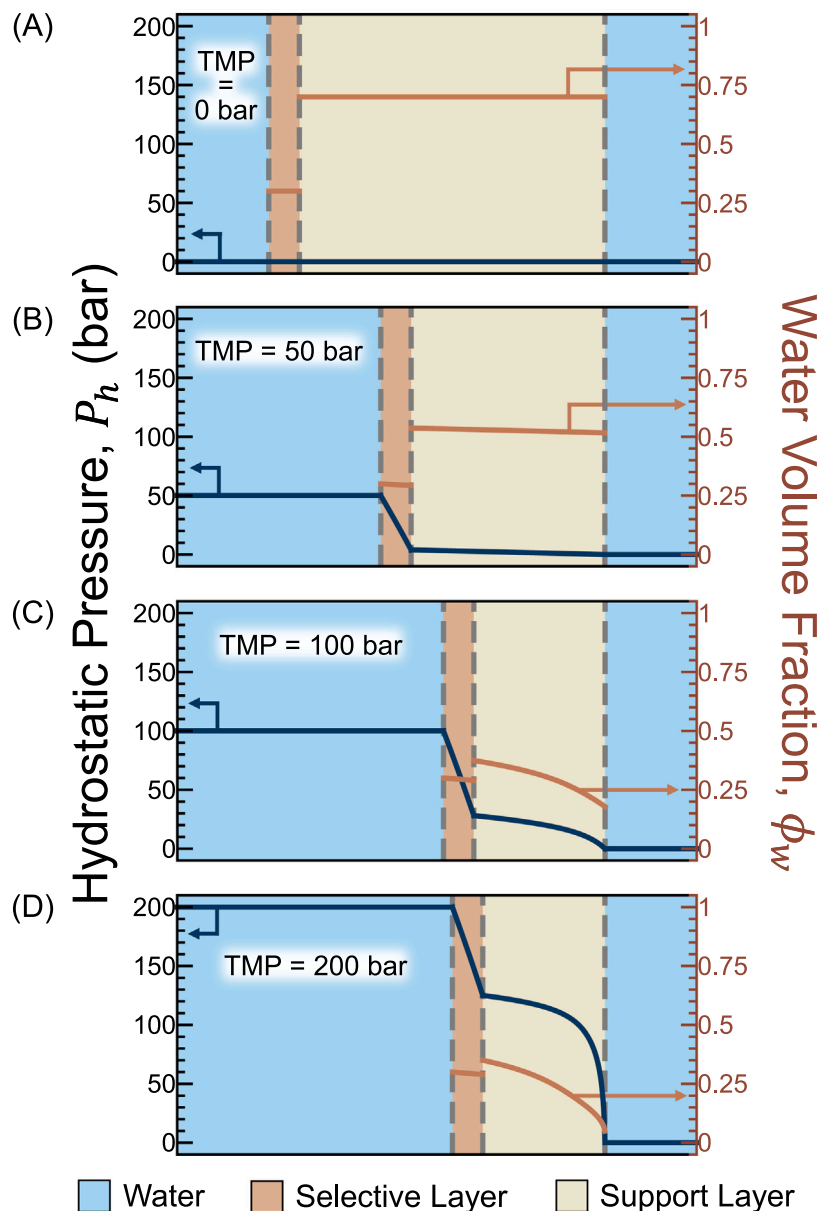


Fig. 7. The true spatial distribution of hydrostatic pressure and porosity (water volume fraction) in our TFC membrane at transmembrane pressure (TMP) of (A) 0 bar, (B) 50 bar, (C) 100 bar, and (D) 200 bar. Note that our selective and support layers are not drawn to scale due to the large disparity in thickness between them (see Table S1). However, the fractional change in thickness of both layers upon TMP application is determined exactly based on Eqs. (13) and (14).

layer. As a result, a noticeable reduction in the support layer thickness and porosity is observed compared to its pristine state, as visualized in Figs. 7(A) and (B).

Notably, our framework does not explicitly account for changes in effective pore size during compaction, which we address for each layer as follows. For the selective layer, Davenport et al. [13] found no significant changes in free volume element size using positron annihilation lifetime spectroscopy (PALS), consistent with our selective layer results in Figs. 7(B)–(D), which indicate minimal changes in porosity and limited compaction. For the support layer, Davenport et al. [13] conducted liquid-liquid porosimetry studies under TMPs of up to 150 bar, revealing consistent pore size distribution (PSD) profiles. In its pristine state, the support layer exhibited a slightly positively skewed, binomial-like PSD. As TMP increased, the height of the PSD

decreased, but its overall shape remained similar. Additionally, the dominant pore size range, approximately 5–10 nm, remained relatively constant across all TMPs tested. These findings suggest that support layer compaction – microscopically linked to a reduction in its porosity – is primarily driven by pore collapse mechanisms that reduce the number of pores rather than their individual sizes (see illustration in SI: Fig. S9). This justifies our constant support layer pore size assumption, allowing us to conveniently quantify changes in support layer porosity during membrane compaction.

When $TMP > 100$ bar, the porosity and hydrostatic pressure of the support layer exhibit non-linear distributions, in contrast to the trends observed in the selective layer. As illustrated in Figs. 7(C) and (D), porosity and hydrostatic pressure transition minimally/linearly across the selective layer, while these quantities decrease more significantly

in the support layer, particularly near the support/permeate interface. This trend qualitatively aligns with experimental [23,67] and theoretical [22–28] findings, suggesting the presence of a more pronounced compacting stress at this interface. In this TMP regime, in addition to the stress transmitted by the selective layer, significant compacting stress arises within the densified support layer itself (see Figs. S13/14 (C) and (D)). The compaction increases toward the support/permeate interface, causing larger local shrinkage and lower local water permeability. Concurrently, to maintain water flow, a higher hydrostatic pressure gradient is necessary near the support/permeate interface, where water permeability declines greatly.

Importantly, the pronounced local compaction near the support/permeate interface highlights how high-pressure reverse osmosis (HPRO) processes inherently require substantial pressure intensity from the feed side. For instance, at TMP = 200 bar, the compaction-induced pressure drop near the support/permeate interface accounts for approximately 50% of the total pressure drop across the TFC membrane. This significantly contributes to the support layer transport resistance (see Fig. 6) under operating conditions typical of HPRO systems (TMP $\gtrsim$ 100 bar [1]), ultimately leading to the plateauing flux regime in Fig. 2. Nevertheless, it is worth noting that HPRO remains a promising technology for desalinating hypersaline brine, as it achieves higher water recovery compared to conventional RO and is more energy-efficient than thermal desalination methods [1,68,69].

To enhance the pressure resistance of TFC membranes, a stiffer support layer can be employed, though this often results in a less porous membrane [13,38]. Therefore, it is essential to achieve an optimal balance in the support layer's poromechanical properties, specifically, between pore size/porosity and mechanical compressibility. This balance enables an effective trade-off between water permeance and compaction resistance, ultimately maximizing HPRO membrane performance under extreme applied pressures. Based on Figs. 7(C) and (D), we suggest a multilayered support layer with increasing stiffness toward the permeate side for pressure-resistant membranes. Particularly, a highly porous, less compressible sublayer can be placed near the selective/support layer interface, while stiffer, less porous sublayers can be used further away to counteract the elevated compacting stress.

To this end, we have demonstrated how pore-flow-driven water transport impacts the membrane morphology (Fig. 7), which links to the permeance reduction (Figs. 3 and 4) and flux-limiting behavior (Fig. 2). To provide further context between different theoretical perspectives, we compare our pore-flow (PF)-based framework with the solution-diffusion (SD) theory in describing the permeation performance of pressure-driven desalination membranes. These two frameworks describe the driving force of water transport in membranes with distinct, mutually exclusive emphases: SD focuses on sorption, followed by diffusion down concentration gradients, while PF emphasizes pressure gradients and viscous flow through relatively distinct and interconnected pores [46,70,71]. While both models can quantify the plateauing flux behavior [70], the underlying mechanisms responsible for this behavior fundamentally differ.

In the SD model, water transport driven by concentration gradients assumes uniform pressure within the membrane, treating the membranes similarly to liquids in how they transmit pressure. To account for the pressure difference between feed and permeate solutions, the model assumes an abrupt hydrostatic pressure drop at the membrane/permeate interface, similar to a piston mechanism [72]—an assumption inconsistent with the behavior of our system. The model assumes chemical equilibrium at the membrane/permeate interface to facilitate this abrupt pressure change, introducing an exponential factor in the expression for water flux [2,70]. Specifically, the SD water flux (#/m²/s) is written as $J_{SD} = D_w \mathcal{K}_w \bar{C}_w \left[1 - \exp\left(-\beta v_w \Delta P_{h,TFC}\right) \right] / L$, where D_w , $\mathcal{K}_w$, $\bar{C}_w$, β and L represent the water diffusivity, water partitioning coefficient in the membrane, bulk water concentration, inverse thermal energy ($(k_B T)^{-1}$) and membrane length, respectively.

In the limit of small TMP, the SD water flux can be approximated as $J_{SD} \approx D_w \mathcal{K}_w \bar{C}_w \left(\beta v_w \Delta P_{h,TFC} \right) / L$, revealing a linear relationship between water flux (J) and TMP ($\Delta P_{h,TFC}$). This behavior resembles the selective layer-controlled regime described in our framework. In the context of SD framework, the linear flux-pressure regime indicates that the water concentration on the membrane's permeate side decreases proportionally with applied pressure. This reduction establishes a concentration gradient, which drives water flow across the membrane. However, with further increment in TMP, the water concentration on the membrane's permeate side would plummet exponentially, effectively reaching zero in the high TMP regime. In this regime, the driving force of the SD framework – the concentration gradient – would reach its maximum limit. Consequently, beyond the linear regime, SD theory predicts that the water flux would deviate from linearity and asymptotically approach $D_w \mathcal{K}_w \bar{C}_w / L$. This behavior is numerically analogous to the support layer-controlled regime illustrated in Fig. 2.

Realistically, the theoretical aspects of the SD model used to describe the plateauing flux particularly the abrupt pressure change at the permeate interface — lack a solid physical foundation. This is because the membranes, in principle, behave more like solids than liquids, consuming pressure while TMP is transmitted across the membranes [30], unlike liquids. In contrast, our PF-based poromechanics framework attributes the flux-limiting behavior to flow-induced compaction, particularly in the support layer. This compaction significantly alters the local porosity, creating substantial resistance to water transport. We can theoretically illustrate this using Eq. (1), where the water flux is locally dependent on two factors: the pressure gradient ($\partial P_h / \partial X$), and Darcy's permeability (k). In the low TMP regime, the TFC membrane remains relatively uncompacted, resulting in an approximately constant k . Consequently, the flux increases proportionally with the pressure gradient, and the slope of the flux-TMP curve is governed by the dense selective layer. Conversely, under high TMP conditions, k within the support layer decreases significantly due to compaction-induced porosity reduction. In response, our model self-consistently predicts a larger pressure gradient to drive water transport. However, this increasing pressure gradient also elevates the compacting stress (see SI: Figs. S13 and S14), leading to an effective cancellation between k and $\partial P_h / \partial X$. As a result, further increases in TMP fail to produce a corresponding increase in water flux, leading to convergence and upper-limit water flux behavior.

Interestingly, at high TMP, the support layer shows only a slight reduction in thickness with increasing pressure, as seen by comparing the membrane lengths at TMP = 100 and 200 bar in Figs. 7(C) and 7(D), respectively. This behavior is attributed to strain-hardening, where the support layer densifies under severe compaction with the collapse of voids. Next, we will quantify the local and cumulative changes in the support layer thickness due to flow-driven compaction to explore this further.

3.4. Strain response of TFC support membrane under flow-induced compaction pressure

The previous subsection shows that the support layer compacts much more than the selective layer. To investigate this further, we quantify the change in the support layer's length during compaction by plotting the global engineering stretch ratio $L_{sup}^{comp} / L_{sup}^o$ against TMP from 0 to 200 bar, where L_{sup}^o and L_{sup}^{comp} represent the initial and compacted support layer lengths, respectively.

Fig. 8(B) compares our stretch $L_{sup}^{comp} / L_{sup}^o$ (dark blue line) with the experimental data from Davenport et al. (light blue hollow circles) [13]. We note that the experimental stretch characterizes only the irreversible deformation of the membrane, which may differ from the envisaged behavior captured by our framework. When pressure is applied to the support layer, it exhibits both reversible and irreversible strains. The reversible strain allows for partial or full recovery of the

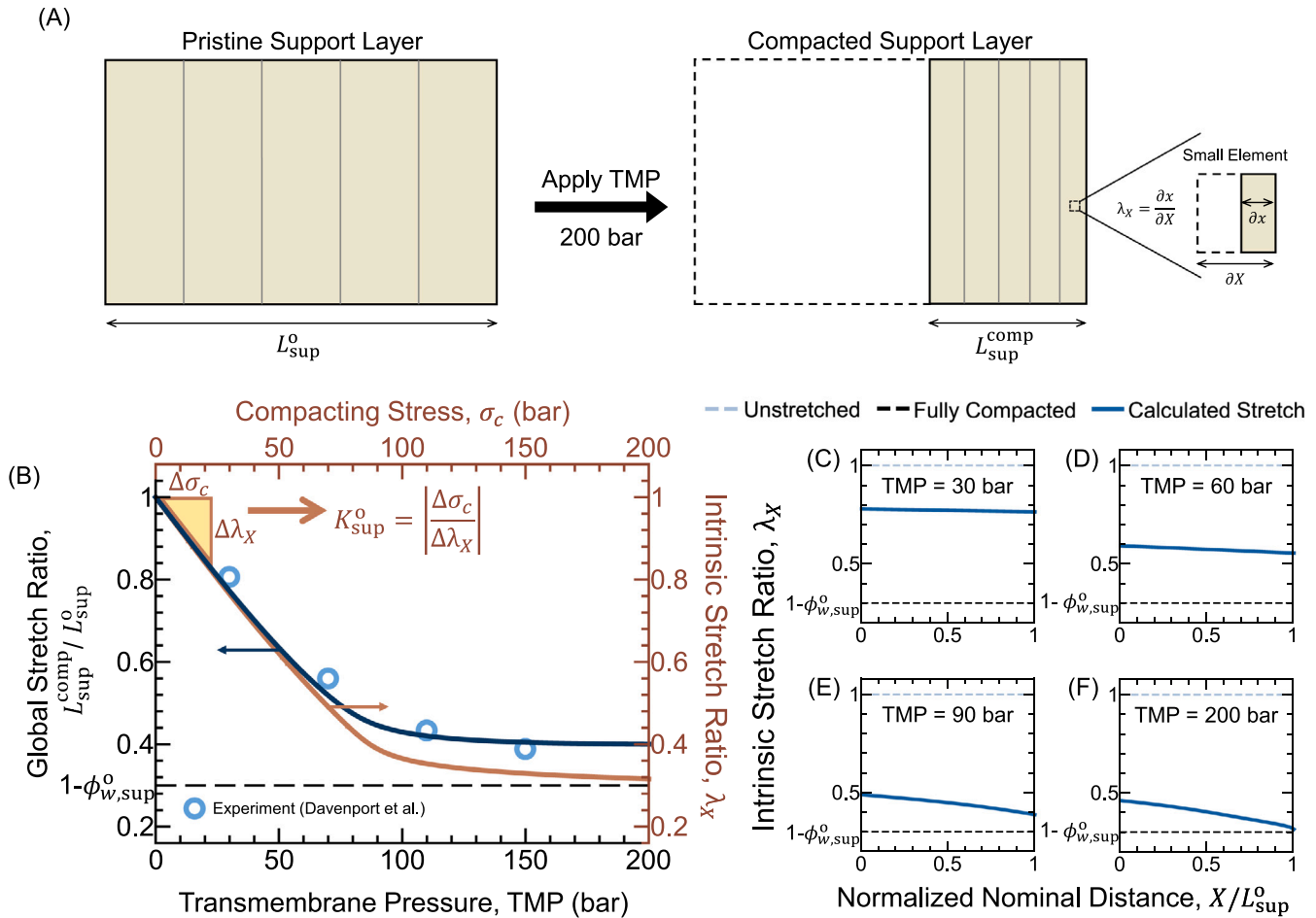


Fig. 8. Detailed depiction of how our TFC support layer mechanically responds to transmembrane pressure (TMP). (A) The visualization of the internally sectioned support layer before and after exposing the TFC membrane to 200 bar TMP (compacted thickness of each sublayer is drawn to scale). (B) The global stretch ratio (dark blue line), compared with experimental post-compaction results [13] (light blue hollow circles), as a function of TMP. In (B), the intrinsic stretch ratio is also plotted as a function of compacting stress for comparative purposes, where the initial stretch-stress ratio is governed by the pristine state compressive modulus, K_{sup}^o (as indicated by the slope triangle). For a more thorough inspection, the nominal distribution of our support layer stretch ratio is presented for TMP of (C) 30 bar, (D) 60 bar, (E) 90 bar, and (F) 200 bar. It is worth noting that in (C)-(F), we have marked two reference stretch levels: the unstretched configuration ($\lambda_x = 1$) in light blue dotted line and the fully-compacted configuration ($\lambda_x = 1 - \phi_{w,sup}^o$) in black dotted line. These references are provided to help readers in gauging the degree of compaction across the support layer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

membrane's thickness upon pressure release, while the irreversible strain accounts for permanent deformation. Typically, ex-situ methods, such as scanning electron microscopy (SEM), measure membrane compaction but capture only irreversible, permanent deformation, as reversible strain diminishes upon pressure unloading due to similar timescales. Consequently, the reversible strain, which is critical to validate our framework, remains unobservable.

Quantifying reversible alongside irreversible strain requires real-time in-situ measurements of membrane compaction, which have not yet been reported over a wide range of applied pressures. Ding et al. [45] reported more than 50% thickness recovery in reverse osmosis (SP475) and nanofiltration (NF270) membranes through creep-recovery and hysteresis experiments. However, their TMPs (< 20 bar) were much lower than those investigated in this study. Experimental data on TFC membrane compaction beyond this pressure range, particularly for the support layer, is limited to post-compaction measurements. These measurements capture only irreversible deformation but still provide valuable insights [11,13,15,16,61]. Davenport et al. [13] and Lim et al. [11] observed only slight recovery in water permeability when

TFC membranes, compacted at pressures up to 150 and 200 bar, respectively, were re-operated at lower pressures. This suggests diminished post-compaction recovery and the dominance of irreversible strain at higher pressures. Therefore, in the absence of comprehensive real-time compaction measurements, our post-compaction measurement-validated L_{sup}^{comp}/L_{sup}^o offers a practical estimate for the upper limit performance of the TFC membrane, as the contribution of reversible strain is negligible under high-pressure conditions.

With the global stretch L_{sup}^{comp}/L_{sup}^o , Fig. 8(B) also presents the intrinsic stretch ratio, λ_x as a function of compacting stress, obtained from Eq. (6). Here, (i) λ_x characterizes the local stretch in response to localized compacting stress, defined as $\lambda_x = \partial x / \partial X$; (ii) L_{sup}^{comp}/L_{sup}^o represents the global stretch of the support layer. Mathematically, the compacted support layer length, L_{sup}^{comp} , is calculated based on Eqs. (13) and (14), expressed as $L_{sup}^{comp} = \int_{X_{sel/sup}}^{X_{sup/perm}} \lambda_x dX$, where $X_{sel/sup}$ and $X_{sup/perm}$ are the nominal coordinates at selective/support and support/permeate interfaces, respectively.

In the low-pressure regime, both the local (λ_x) and the global (L_{sup}^{comp}/L_{sup}^o) stretches decrease linearly with applied pressure. As illustrated by the slope triangle in Fig. 8(B), this behavior is governed

by the compressive modulus of the pristine support layer, K_{sup}^0 , where the support layer comprises a polymer-water mixture. We estimate a compressive modulus K_{sup}^0 of 12.5 MPa, which is an order of magnitude lower than the tensile modulus of commercial support membranes [11, 73]. Nevertheless, our K_{sup}^0 value aligns with experimental observations of TFC membrane compaction reported by Ding et al. [45], who discussed the difference between the compressive and tensile strains of TFC membranes in great detail.

In the high-pressure zone, strain-hardening causes the compressive stress to diverge exponentially as the membrane porosity decreases (see SI: Section S4), indicating an increase in membrane stiffness. This contrasts with tensile tests, where membranes typically (rubbery) yield and (glassy) fracture [73]. The strain-hardening behavior in our support layer stems from mechanisms similar to those in cellular materials under compression (e.g., foams), where bending and buckling of cell walls [74–76] cause void collapse [13], leading to densification (see Figs. 7(C) and (D)) and reduced compressibility due to tighter polymer packing. To relate, Aghajani et al. [56] found that pre-compacted membranes resist further compaction, showing less reduction in water permeability, which points to strain-hardening. Here, we capture this strain-hardening behavior phenomenologically by parametrically adjusting the coefficients β_1 and β_2 (see Eq. (6)), which control the deviation of the stress–stretch relationship from linear behavior. As a limiting case, when λ_X reaches $1 - \phi_{w,\text{sup}}^0$, the TMP increases asymptotically, as expected (see Fig. 8(B)).

With low TMP, the support layer deforms uniformly, as indicated by the nearly constant λ_X in Fig. 8(C). Correspondingly, the experimental stretch $L_{\text{sup}}^{\text{comp}}/L_{\text{sup}}^0$ coincides with its local intrinsic λ_X one in Fig. 8(B). We can demonstrate this correlation mathematically by writing $L_{\text{sup}}^{\text{comp}} = \lambda_X \int_{X_{\text{sel/sup}}}^{X_{\text{sup/perm}}} dX = \lambda_X L_{\text{sup}}^0$, which yields $L_{\text{sup}}^{\text{comp}}/L_{\text{sup}}^0 = \lambda_X$. When TMP is high, λ_X declines toward the support/permeate interface (see Figs. 8(D)–(F)), similar to how the porosity distribution changes (see Fig. 7). As λ_X exhibits a decreasing non-uniform distribution under this applied pressure, we have $\int_{X_{\text{sel/sup}}}^{X_{\text{sup/perm}}} \lambda_X dX \neq \lambda_X L_{\text{sup}}^0$, which explains the numerical difference between the (global) experimental and (local) intrinsic stretch ratios plotted in Fig. 8(B).

We further explain the local compaction trends in the support layer at various applied pressures by dividing it into five sublayers, as illustrated in Fig. 8(A), inspired by previous stacked membrane water permeation experiments [77–80]. At TMP = 200 bar, there is no visible difference in the macroscopic thickness between the five compacted sublayers (see Fig. 8(A)), despite the variations in λ_X distribution (see Fig. 8(F)). This is because, microscopically, the sublayers are already near or at their polymer packing limit.

The trend in Fig. 8(A) slightly differs from experimental studies [23, 77–80] and poromechanics models [22,23,26,27,81], which examine flow-induced compaction. As a comparison, these studies suggest minimal compaction near the feed solution, with porosity and stretch ratios close to the uncompacted state, whereas our *bilayer* results show severe compaction across the *entire* support layer, as evidenced by significant drops in porosity and stretch in Figs. 7 and 8. Nonetheless, both our work and other poromechanics frameworks reveal non-uniformly distributed stretch within the membrane, indicating the presence of a pressure gradient—a contrast to the uniform pressure distribution predicted by the SD model.

To further clarify our support layer compaction trend in Figs. 8(A), 7(B) shows that under low applied pressure conditions (~ 50 bar TMP), the selective layer transmits pressure to the support layer due to pressure continuity, resulting in noticeable and relatively uniform compaction. On one hand, this compaction triggers strain-hardening in the support layer, which increasingly resists further compression as the TMP rises, limiting the variation of morphological change across the support layer. On the other hand, attributed to compaction-induced transport resistance, non-linear hydrostatic pressure gradient arises within the support membrane, which tends to compact the region near

the support/permeate interface more. Collectively, at 200 bar TMP, the interplay of strain-hardening and applied pressure leads to a non-linear stretch distribution, but less pronounced compared to the hydrostatic pressure one (see Figs. 5(D) and 7(D)), resulting in the approximately constant-thickness compacted sublayers observed in Fig. 8(A).

4. Conclusion and implications

This study introduces a poromechanics framework that integrates non-linear mechanics with non-ideal pore-flow transport to unravel the complex interplay between pressure-driven (i) compaction and (ii) water transport in TFC membranes. By explicitly accounting for the distinct structural, mechanical, and transport properties of the selective and support layers, the framework realistically highlights their significant combined impact on the overall water permeation performance of TFC membranes.

Based on their individual permeance values (see Fig. 3), we demonstrate that the dense selective layer governs water transport at low TMP, whereas the support layer governs transport at high TMP due to its lower permeance. In the low TMP regime, as TMP increases, the permeation-controlling selective layer maintains a linear hydrostatic pressure distribution, which shifts upward with minimal change in its gradient. In contrast, in the high TMP regime, the support layer undergoes significant compaction (see Figs. 7 and 8), leading to increased transport resistance. This compaction consumes a substantial portion of the TMP, causing the support layer to exhibit a steep, non-linear hydrostatic pressure distribution. Consequently, at high TMP levels, the TFC membrane transitions from a selective-layer-controlled water transport system to one limited by the support layer. These findings suggest that while current MD simulations excel in capturing atomistic details of water transport through the selective layer of a TFC membrane, they fall short of providing a complete understanding of the transport-limiting mechanisms without explicitly simulating the support layer. This limitation is particularly pronounced under high TMP operating conditions.

Under flow-induced compaction, particularly at high TMP levels, experiments demonstrate that TFC support membranes exhibit strain-hardening behavior, characterized by a steeper stress–stretch relationship where stress increases more rapidly than stretch [11,13,61], consistent with our prediction of support layer compaction. Although this behavior resembles the prediction of Flory–Rehner (FR) theory, which suggests an exponential increase in compacting stress with compressive stretch, FR theory significantly overestimates water flux at high TMP, predicting values several times greater than both our theoretical results and experimental observations. This discrepancy stems from the thermodynamic foundation of FR theory, which predicts unrealistically high compacting stress in fully-compacted membranes with very low water content due to the unphysical enlargement of osmotic pressure. In contrast, our stress–stretch relationship realistically captures the membrane deformation across the entire range, from the pristine state ($\lambda_X = 1$) to the fully-compacted one ($\lambda_X = 1 - \phi_{w,\text{sup}}^0$), by transitioning from the mechanical response of a hydrated membrane (water-polymer mixture) to that of a solid material (bulk polymer), enabling the capture of (i) selective and (ii) support layer-limited flux-pressure regimes corresponding to low and high TMPs, respectively, as illustrated in Fig. 2.

In summary, this study elucidates the flow-induced compaction of a bilayer TFC membrane by quantifying the spatially dependent variables, such as the pressure and porosity distributions, which are challenging to measure experimentally. These findings underscore the individual influence of the support and selective layers, allowing us to understand further the intricate coupling between deformation and water permeation within the TFC membrane. However, it is important to note that the current model does not account for the influence of other membrane physicochemical properties, such as hydrophilicity and composition, which could alter the membrane behavior [7,40,60].

In future work, expanding the model to incorporate these factors, along with multi-ionic salt transport [82–85] to mimic seawater/riverwater conditions, could offer a more comprehensive understanding of TFC membrane behavior under high pressure.

CRedit authorship contribution statement

Ian Keen Koo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Meng Nan Chong:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **K.B. Goh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124023>.

Data availability

Data will be made available on request.

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